Chapter 4. Qualitative Theory of Differential Equations
Sections 4.1, 4.2, 4.3

1. Consider the differential equation

$$\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x}) \tag{1}$$

where $\mathbf{x} = \begin{bmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{bmatrix}$ and $\mathbf{f}(t, \mathbf{x}) = \begin{bmatrix} f_1(t, x_1, \cdots, x_n) \\ \vdots \\ f_n(t, x_1, \cdots, x_n) \end{bmatrix}$.

- $\mathbf{x}^0$ is called a *equilibrium value* of (1) if and only if

$$\mathbf{f}(t, \mathbf{x}^0) \equiv 0.$$

- Definition of *stability* of a solution $\mathbf{x} = \phi(t)$ of (1), see Page 378.
- Definition of *asymptotical stability* of a solution $\mathbf{x} = \phi(t)$ of (1), see Page 381.

2. Consider the homogeneous linear differential equation

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} \tag{2}$$

where $\mathbf{A}$ is an $n \times n$ matrix.

- Every solution of (2) is *asympotically stable* if all eigenvalues of $\mathbf{A}$ have negative real part.
- Every solution of (2) is *unstable* if at least one eigenvalue of $\mathbf{A}$ has positive real part.
- Suppose that all eigenvalues of $\mathbf{A}$ have real part ≤ 0 and $\lambda_1, \cdots \lambda_l$ have zero real part with multiplicity k_j. Then, every solution of (2) is *stable* if $\mathbf{A}$ has k_j linearly independent eigenvectors for each eigenvalue of $\lambda_1, \cdots, \lambda_l$. Otherwise, every solution of (2) is *unstable*.

3. The stability of any solution of the nonhomogeneous equation

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{f}(t) \tag{3}$$

is equivalent to the stability of the solution of (2).

4. Consider the nonlinear differential equation

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{g}(\mathbf{x}) \tag{4}$$

where $\mathbf{g}(\mathbf{x}) = \begin{bmatrix} g_1(\mathbf{x}) \\ \vdots \\ g_n(\mathbf{x}) \end{bmatrix}$ and assume that $\dfrac{\mathbf{g}(\mathbf{x})}{\max\{x_1, \cdots, x_n\}}$ is a continuous function which vanishes for $\mathbf{x} = \mathbf{0}$. Clearly, $\mathbf{x}(t) \equiv 0$ is an equilibrium solution of (4). Then

- The equilibrium solution $\mathbf{x}(t) \equiv 0$ of (4) is *asympotically stable* if all eigenvalues of $\mathbf{A}$ have negative real part.
- The equilibrium solution $\mathbf{x}(t) \equiv 0$ of (4) is *unstable* if at least one eigenvalue of $\mathbf{A}$ has positive real part.
- Suppose that all eigenvalues of $\mathbf{A}$ have real part ≤ 0 and $\lambda_1, \cdots \lambda_l$ have zero real part with $l \geq 1$. Then, the stability of The equilibrium solution $\mathbf{x}(t) \equiv 0$ of (4) cannot be determined.

5. **Algorithm** for determining the stability of the equilibrium solution $\mathbf{x}(t) \equiv \mathbf{x}^0$ of $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$:

   1. Set $\mathbf{z} = \mathbf{x} - \mathbf{x}^0$.
   2. Write $\mathbf{f}(\mathbf{x}^0 + \mathbf{z})$ in the form $\mathbf{A}\mathbf{z} + \mathbf{g}(\mathbf{z})$ where $\mathbf{g}(\mathbf{z})$ is a vector-value polynomial in $z_1, \cdots, z_n$ beginning with terms of order two or higher.
      - In fact, we have that

$$\mathbf{A} = \begin{bmatrix} \dfrac{\partial f_1}{\partial x_1}(\mathbf{x}^0) & \cdots & \dfrac{\partial f_1}{\partial x_n}(\mathbf{x}^0) \\ \vdots & \vdots & \vdots \\ \dfrac{\partial f_n}{\partial x_1}(\mathbf{x}^0) & \cdots & \dfrac{\partial f_n}{\partial x_n}(\mathbf{x}^0) \end{bmatrix}$$

   3. Compute all eigenvalues of $\mathbf{A}$.
   4. If all eigenvalues of $\mathbf{A}$ have negative real part, then $\mathbf{x}(t) \equiv \mathbf{x}^0$ is *asympotically stable*. If at least one eigenvalue of $\mathbf{A}$ has positive real part, then $\mathbf{x}(t) \equiv \mathbf{x}^0$ is *unstable*.